



PIZZA SALES

BY ADIBA SABRI

DISCOUNT · DISCOUNT · DISCOUNT · DISCOUNT · DISCOUNT

25%

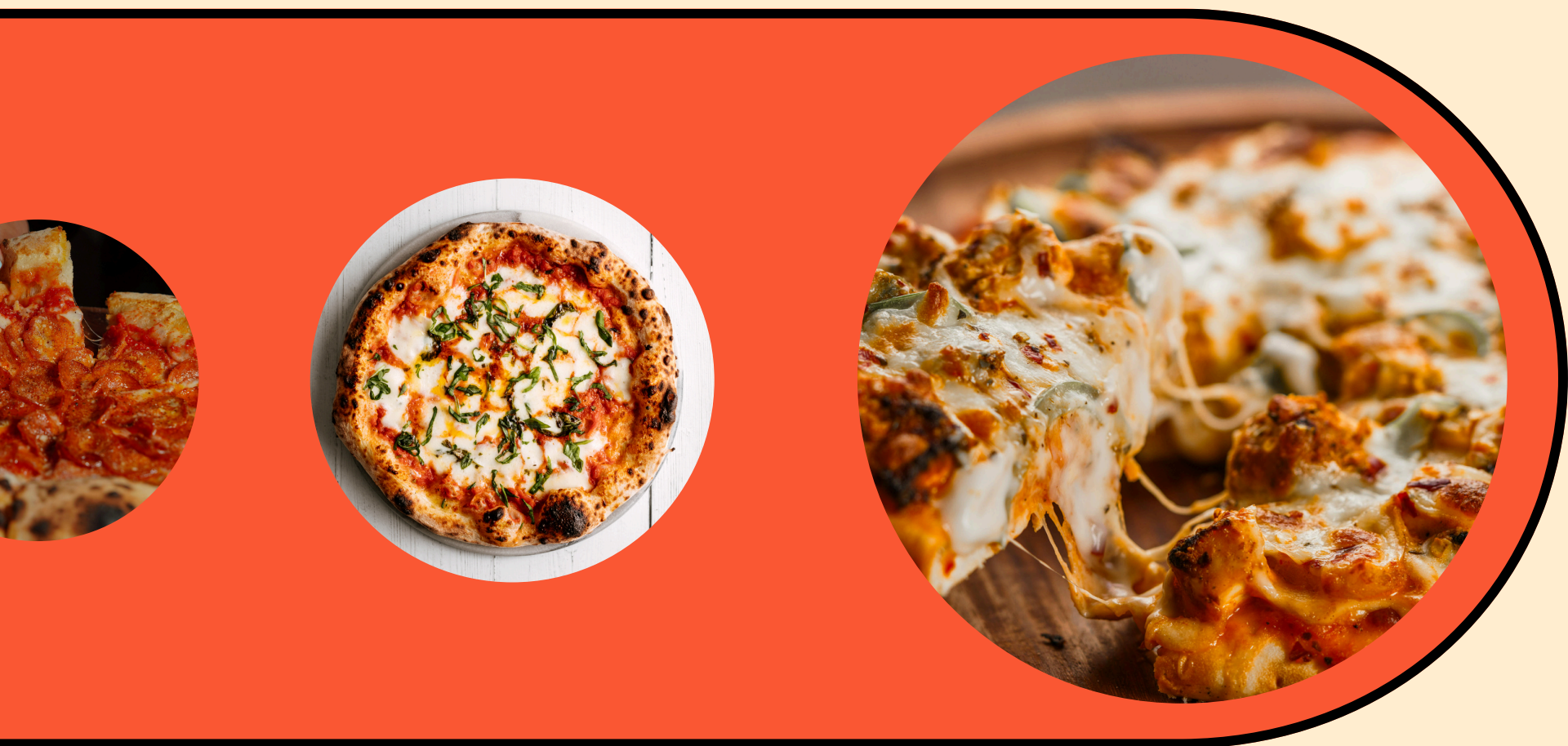


ABOUT ME

Hi ! I am an aspiring data analyst with a strong interest in transforming raw data into meaningful insights . I am currently developing my skills in SQL, Python, Excel, and Power BI to solve real world bussiness problems.



PROJECT OVERVIEW



This project focuses on analyzing pizza sales data analysis using sql to explore sales performances , customer ordering patterns, and product demand.

The project involves aggregation , joins, filtering , subqueries , common table expressions CTEs and window function etc .

TOOL AND TECHNOLOGY USED

- . sql
- . postgresql

Retrieve the total number of orders placed.

```
select count(order_id) as total_orders from orders;
```

	total_orders bigint 
1	21350

Calculate the total revenue generated from pizza sales.

```
select  
round(sum(od.quantity*p.price),2) as total_revenue  
from order_details as od join pizzas as p on od.pizza_id = p.pizza_id;
```

	total_revenue numeric 
1	817860.05

Identify the highest-priced pizza.

```
select  
pt.name,p.price  
from pizza_types as pt join pizzas as p on pt.pizza_type_id = p.pizza_type_id  
order by p.price desc  
limit 1;
```

	name text	price numeric
1	The Greek Piz...	35.95

Identify the most common pizza size ordered.

```
select count(od.order_id) as order_count, pizzas.size
from order_details as od join pizzas on od.pizza_id = pizzas.pizza_id
group by pizzas.size
order by order_count desc
```

	order_count 	size 
1	18526	L
2	15385	M
3	14137	S
4	544	XL
5	28	XXL

Join the necessary tables to find the total quantity of each pizza category ordered.

```
select sum(od.quantity),pt.category
from order_details as od join pizzas as p on od.pizza_id = p.pizza_id
join pizza_types as pt on pt.pizza_type_id = p.pizza_type_id
group by category;
```

	sum bigint 	category text 
1	11987	Supreme
2	11050	Chicken
3	14888	Classic
4	11649	Veggie

Determine the distribution of orders by hour of the day.

```
select count(order_id) as order_count, extract(hour from order_time) as hours
from orders
group by hours
order by order_count desc;
```

	order_count  bigint	hours  numeric
	2520	12
	2455	13
	2399	18
	2336	17
	2009	19
	1920	16
	1642	20
	1472	14
	1468	15
	1231	11
	1198	21
	663	22
	28	23
	8	10
	1	9

Join relevant tables to find the category-wise distribution of pizzas.

```
select count(name), category  
from pizza_types  
group by category;
```

count bigint	category text
9	Supreme
6	Chicken
8	Classic
9	Veggie

Group the orders by date and calculate the average number of pizzas ordered per day.

```
select round(avg(total_quantity),0) as average_pizzas_ordered_per_day
from (select order_date, sum(quantity) as total_quantity
from orders as o join order_details as od on o.order_id = od.order_id
group by order_date);
```

average_pizzas_ordered_per_day 
138

Determine the top 3 most ordered pizza types based on revenue.

```
select pt.name, sum((p.price)*(od.quantity)) as revenue
from pizza_types as pt join pizzas as p on pt.pizza_type_id = p.pizza_type_id
join order_details as od on od.pizza_id = p.pizza_id
group by pt.name
order by revenue desc
limit 3
```

name text	revenue numeric
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Piz...	42768.00
The California Chicken Piz...	41409.50

Calculate the percentage contribution of each pizza type to total revenue.

```
with total_sales as (  
  select sum(od.quantity* p.price) as t_sales  
  from order_details as od join pizzas as p on od.pizza_id = p.pizza_id  
)  
select pt.category,round(sum(od.quantity*p.price)*100/total_sales.t_sales,2) as total_revenue  
from pizza_types as pt join pizzas as p on pt.pizza_type_id = p.pizza_type_id  
join order_details as od on od.pizza_id = p.pizza_id  
cross join total_sales  
group by pt.category,total_sales.t_sales  
order by total_revenue desc
```

category 	total_revenue 
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

Analyze the cumulative revenue generated over time.

```
select order_date , sum(revenue) over(order by order_date) as cumulative_rev
from
(select orders.order_date,sum(order_details.quantity*pizzas.price) as revenue
from orders join order_details on orders.order_id =order_details.order_id
join pizzas ON pizzas.pizza_id = order_details.pizza_id
group by orders.order_date
order by orders.order_date) as sales
```

order_date date	cumulative_rev numeric
2015-01-01	2713.85
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.60
2015-01-05	11929.55
2015-01-06	14358.50
2015-01-07	16560.70

Determine the top 3 most ordered pizza types based on revenue for each pizza category

```
select category, name, revenue, rank() over(partition by category order by revenue desc) as rev_rank
from(
select pizza_types.category, pizza_types.name, sum(order_details.quantity*pizzas.price) as revenue
from pizza_types join pizzas on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name
order by revenue desc)
```

category text	name text	revenue numeric	rev_rank bigint
Chicken	The Thai Chicken Pizza	43434.25	1
Chicken	The Barbecue Chicken Pizza	42768.00	2
Chicken	The California Chicken Pizza	41409.50	3
Chicken	The Southwest Chicken Pizza	34705.75	4
Chicken	The Chicken Alfredo Pizza	16900.25	5
Chicken	The Chicken Pesto Pizza	16701.75	6
Classic	The Classic Deluxe Pizza	38180.5	1
Classic	The Hawaiian Pizza	32273.25	2
Classic	The Pepperoni Pizza	30161.75	3
Classic	The Greek Pizza	28454.10	4

Classic	The Italian Capocollo Pizza	25094.0	5
Classic	The Napolitana Pizza	24087.0	6
Classic	The Big Meat Pizza	22968	7
Classic	The Pepperoni, Mushroom, and Peppers Pi...	18834.5	8
Supreme	The Spicy Italian Pizza	34831.25	1
Supreme	The Italian Supreme Pizza	33476.75	2
Supreme	The Sicilian Pizza	30940.50	3
Supreme	The Pepper Salami Pizza	25529.00	4
Supreme	The Prosciutto and Arugula Pizza	24193.25	5
Supreme	The Soppressata Pizza	16425.75	6
Supreme	The Calabrese Pizza	15934.25	7
Supreme	The Spinach Supreme Pizza	15277.75	8
Supreme	The Brie Carre Pizza	11588.50	9
Veggie	The Four Cheese Pizza	32265.70	1



THANK YOU